

Quantitative Marketing Research

From Business Questions to Data-Driven Answers

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February 12, 2026

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Imagine You Are the Marketing Manager...

1. **Nike's social media team** notices that their posts with the highest number of impressions seem to get lower engagement rates. Should they stop trying to maximize reach?
2. **Apple's brand manager** wants to know: do posts about Product topics drive more engagement than posts about Pricing or Support?
3. **Coca-Cola vs. Pepsi:** A VP asks – “Is there a real difference in how people engage with our brand versus the competitor, or is it just random noise?”

Why Do We Need Numbers? (Part 1)

- Intuition is valuable but often **wrong** at scale
- Quantitative research gives us three superpowers:
 - **Measure** effect sizes precisely
 - **Compare** alternatives with confidence
 - **Separate** real patterns from noise
- Marketing decisions involve millions of euros

Example: When Intuition Fails

Your Intuition:

"I think negative posts get more likes."

What the Data Shows:

*Positive posts actually get the highest engagement rate (**0.307** vs. **0.262** for positive).*

This is why we need numbers – our intuitions are often wrong.

The Research Process

Business Problem → Research Question → Data Collection → Analysis → Decision

Step	Example
Business Problem	“Our social media engagement is stagnating”
Research Question	“Which factors explain variation in engagement rate?”
Data	12,000 posts across 10 brands and 5 platforms
Analysis	Correlation, regression, hypothesis tests
Decision	Reallocate budget toward high-engagement topics

Variance – How Spread Out Is the Data?

- **Variance** measures how much data points differ from the average
- Low variance = clustered tightly; High variance = spread out
- Formula:

$$\text{Variance} = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

- **Why it matters:** If there is no variance, there is nothing to explain

Correlation – Do Two Variables Move Together?

- Correlation measures the **direction** and **strength** of a linear relationship between two numerical variables
- Ranges from **-1** to **+1**:
 - **+1** = perfect positive (as X goes up, Y goes up)
 - **0** = no linear relationship
 - **-1** = perfect negative (as X goes up, Y goes down)
- Notation: r (sample correlation) or ρ (population correlation)
- Important: correlation measures **LINEAR** relationships only

-1.0	-0.5	0	+0.5	+1.0
Perfect	Moderate	No	Moderate	Perfect
Negative	Negative	Correlation	Positive	Positive

How Is Correlation Calculated?

Pearson correlation coefficient (r):

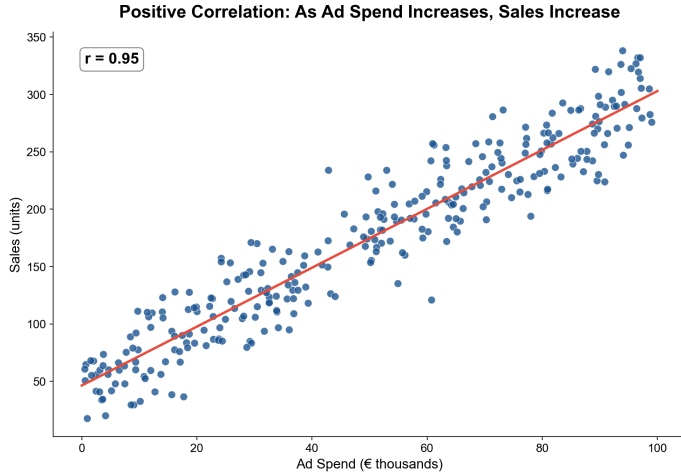
$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \cdot \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

In plain English:

- Measures how much X and Y vary **together** relative to how much they vary **separately**
- Ranges from -1 (perfect negative) to +1 (perfect positive)
- $r = 0$ means no linear relationship

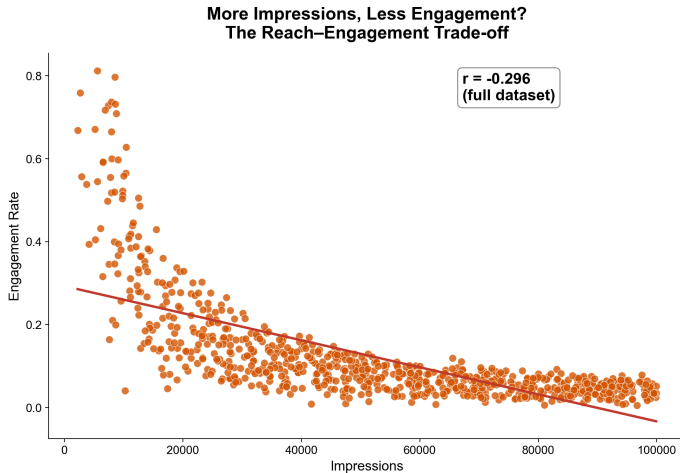
*You don't need to memorize this formula – software calculates it for you.
What matters is understanding what the number means.*

Positive Correlation Example



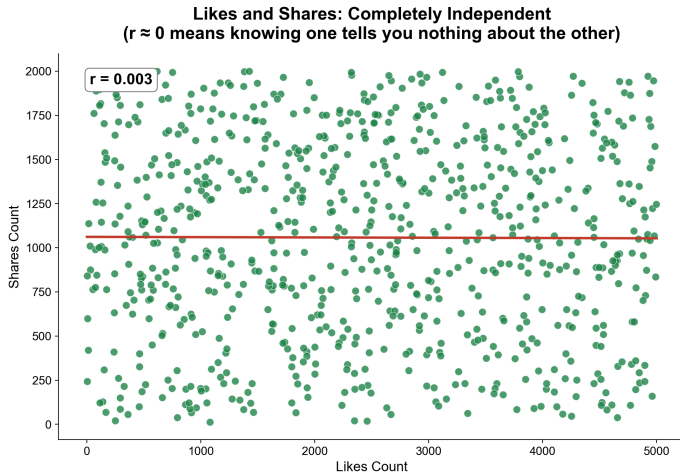
Posts with higher ad spend tend to get higher sales – a strong positive correlation.

Negative Correlation: Impressions and Engagement Rate



Posts that reach **MORE** people tend to have **LOWER** engagement rates ($r = -0.30$).

No Correlation: Likes and Shares



Knowing how many likes a post got tells you **nothing** about how many shares it will get ($r = 0.003$).

Correlation Does Not Mean Causation (Part 1)

- Just because two variables move together does not mean one causes the other
- Three possible explanations for a correlation between X and Y:
 1. **X causes Y** (direct causation)
 2. **Y causes X** (reverse causation)
 3. **Z causes both X and Y** (confounding / spurious correlation)
- Classic example: ice cream sales and drowning rates both go up in summer (confounded by temperature)
- **More hilarious examples:** <https://www.tylervigen.com/spurious-correlations>

Correlation Does Not Mean Causation (Part 2)

In our data: if we find that sentiment_score correlates with engagement_rate, is it because:

1. **Positive sentiment drives engagement?** (X causes Y)
2. **High engagement makes people write more positively?** (Y causes X)
3. **Certain brands/topics produce both positive sentiment AND high engagement?**
(confounder Z)

We need regression with controls to untangle these possibilities.

Controlling for Confounders

- A **control variable** is a factor we hold constant to isolate the relationship we care about
- Example: To study the effect of sentiment on engagement, we need to control for brand and platform
 - Without controls: sentiment and engagement might correlate simply because both are driven by brand
 - With controls: we compare posts from the **same brand, same platform** and ask – among those, does sentiment still matter?
- This is the bridge from correlation to regression

Holding brand constant, does sentiment still predict engagement?

Linear Regression – Finding the Best-Fitting Line

Regression equation:

$$Y = \alpha + \beta X + \varepsilon$$

- α = intercept
- β = slope (key number!)
- ε = error

“Best-fitting” = line closest to all data points

The Slope (Beta) – The Key Number

- The slope β (also called beta) is the most important output of a regression
- Interpretation: “For every 1-unit increase in X , Y changes by β units”
- The sign of β tells direction:
 - $\beta > 0$: positive relationship
 - $\beta < 0$: negative relationship
 - $\beta = 0$: no relationship

“For every one-unit increase in X , Y changes by β units, on average.”

Interpreting Beta: An Example

Example with our data:

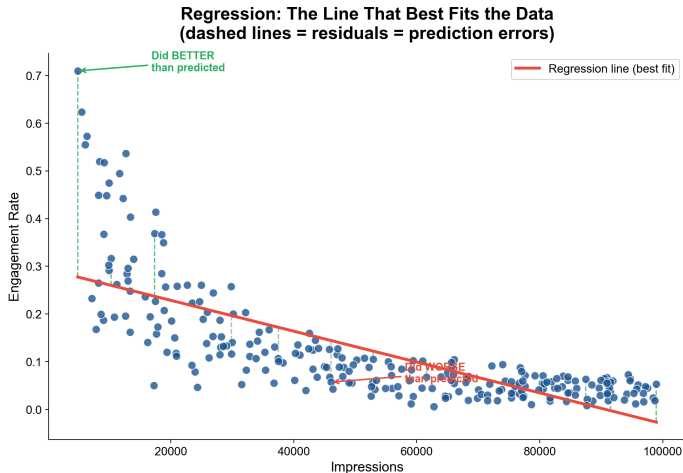
If we regress `engagement_rate` on `sentiment_score`:

$$\beta = 0.042$$

Interpretation: A **1-point increase** in sentiment (e.g., from 0.0 to 1.0) is associated with a **0.042 increase** in engagement rate

This is the “effect size” – what marketers need for resource allocation decisions.

Regression in Action: Sentiment Predicting Engagement Rate



The red line is the regression line. Vertical distances = residuals = prediction errors.

R-Squared – How Much Variance Does the Model Explain?

- R-squared (R^2) measures the proportion of variance in Y explained by the model
- Ranges from **0 to 1**:
 - $R^2 = 0$: the model explains **nothing** (the line is useless)
 - $R^2 = 1$: the model explains **everything** (all points on the line)
 - $R^2 = 0.08$ (for impressions predicting engagement): the model explains **8% of the variance**
- That means **92%** of the variation in engagement rate is due to **OTHER factors**

Multiple Regression – Including Control Variables

- We can add multiple predictors to the model:

$$\text{engagement_rate} = \alpha + \beta_1 \cdot \text{sentiment} + \beta_2 \cdot \text{brand} + \beta_3 \cdot \text{platform} + \varepsilon$$

- Each β coefficient now represents the effect of that variable **holding all others constant**
 - β_1 = effect of sentiment on engagement, **controlling for** brand and platform
- This addresses the confounding problem from earlier
- Categorical variables (brand, platform) are included as “dummy variables” – the software handles this

Each coefficient = the effect of that variable, holding all others constant

What Predicts Engagement Rate? (Multiple Regression Results)

Predictor	Coefficient (β)	Interpretation
sentiment_score	0.03	Positive effect, controlling for all else
Brand: Nike	0.15	Nike's engagement premium vs. baseline
Topic: Product	0.08	Product topic premium vs. baseline
R^2		0.58

The full model explains more variance than any single predictor alone.

Population vs. Sample

- **Population:** All social media posts about these **10 brands (millions)**
- **Sample:** Our dataset of **12,000 posts**
- We use the sample to make **inferences** about the population
- Every estimate has **uncertainty**

The dataset is a window into the full population.

Example: Estimating Average Height in Spain

Research Question: What is the average height of university students in Spain?

What we do:

- We measure **10 students** in this classroom
- We calculate the **average height** of these 10 students
- We use this to **estimate** the population average

Heights (cm): 172, 168, 175, 165, 180, 171, 169, 177, 173, 170

Can we trust this sample of 10 to tell us about millions of students?

Calculating the Confidence Interval

Step 1: Calculate the sample mean

$$\bar{x} = \frac{172 + 168 + 175 + 165 + 180 + 171 + 169 + 177 + 173 + 170}{10} = 172 \text{ cm}$$

Step 2: Calculate the standard deviation

$$s = 4.5 \text{ cm}$$

Step 3: Calculate the standard error

$$SE = \frac{s}{\sqrt{n}} = \frac{4.5}{\sqrt{10}} = 1.4 \text{ cm}$$

Step 4: Build the 95% confidence interval

$$CI_{95\%} = \bar{x} \pm 1.96 \times SE = 172 \pm 1.96 \times 1.4 = 172 \pm 2.7$$

$$CI_{95\%} = [169.3, 174.7] \text{ cm}$$

What Does This Confidence Interval Mean?

Our 95% confidence interval: [169.3, 174.7] cm

Interpretation:

- We are **95% confident** that the **true average height** of Spanish university students is between **169.3 and 174.7 cm**
- If we repeated this experiment **100 times** (measuring 10 different students each time), about **95 of the intervals** would contain the true population mean
- The interval is **wide** (± 2.7 cm) because our sample is **small** ($n = 10$)

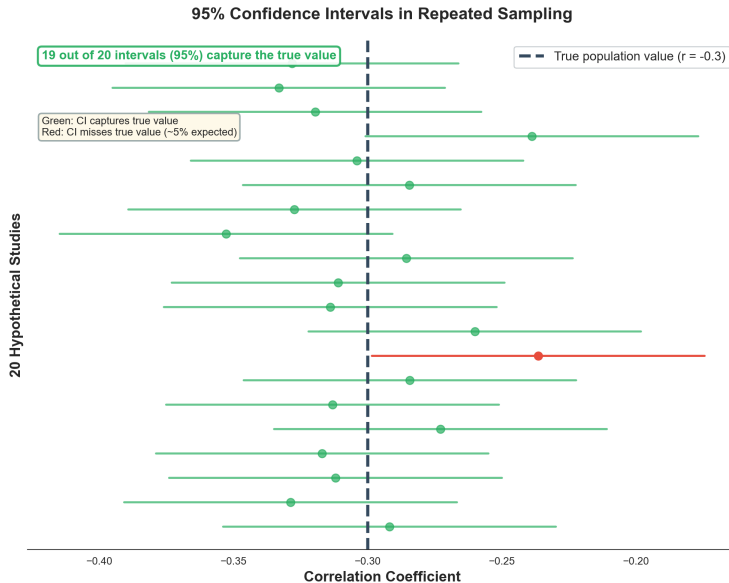
Larger samples $\rightarrow$ narrower intervals $\rightarrow$ more precision

How Certain Are We About Our Estimates?

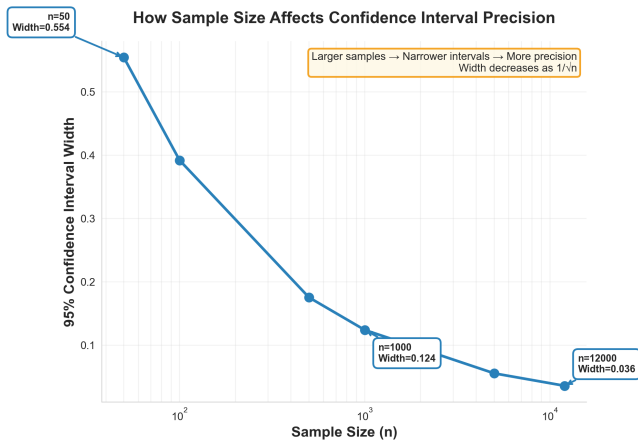
- Every statistic is an **estimate** with **uncertainty**
- **Confidence interval:** “We are 95% confident the true value is between X and Y”
- Narrow interval = high precision; Wide interval = low precision
- Formula: $CI = \text{estimate} \pm 1.96 \times \frac{s}{\sqrt{n}}$
 - s = standard deviation, n = sample size

*If we repeated the study **100 times**,
95 intervals would contain the true value*

Confidence Intervals: Visual Interpretation



Why Sample Size Matters: Precision of Estimates



- Larger sample → narrower CI → more precision
- CI width $\propto 1/\sqrt{n}$

Example: How Much Does Sentiment Affect Engagement?

- From our regression: $\text{engagement_rate} = \alpha + \beta \times \text{sentiment_score} + \dots$
- Point estimate: $\beta = 0.042$
 - Interpretation: A **1-unit** increase in sentiment is associated with a **0.042 increase** in engagement rate
- **95% Confidence Interval: [0.028, 0.056]**
 - Interpretation: We are **95% confident** the true effect is between **0.028 and 0.056**
- Since the interval does **NOT include zero**, we can be confident sentiment has a **real, positive effect**
- This is the visual counterpart to hypothesis testing (which we will cover next)

The Fundamental Problem: Is It Real or Is It Noise?

- We found $\beta = 0.042$ for the effect of sentiment on engagement
- Does this exist in the **population**, or just our **sample**?
- Hypothesis testing answers: **Is this real or random noise?**

Quantifying confidence with a known error rate

Setting Up Hypotheses

Null Hypothesis (H_0): There is **no effect** / no relationship / no difference

- “Sentiment score has no effect on engagement ($\beta = 0$)”
- “There is no difference in engagement between Nike and Adidas”
- “Brand has no effect on engagement”

Alternative Hypothesis (H_1 or H_a): There **is** an effect / relationship / difference

- “Sentiment DOES affect engagement ($\beta \neq 0$)”
- “There IS a difference between Nike and Adidas”
- “Brand DOES affect engagement”

We start by assuming H_0 is true and ask: how likely is our data under that assumption?

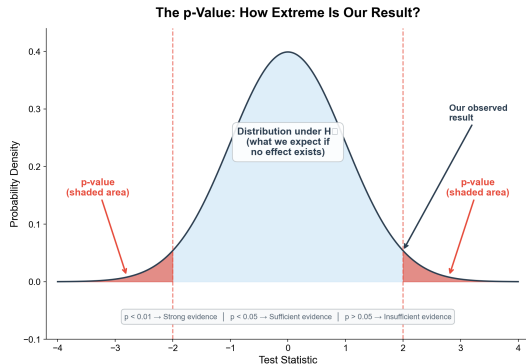
How Hypothesis Testing Works

The logic, step by step:

1. **Assume H_0 is true** (no effect exists)
2. **Ask:** If H_0 were true, how likely is it that we would observe data as extreme as what we got?
3. **Calculate this probability** – this is the **p-value**
4. **Decision rule:**
 - If p-value is very small (typically < 0.05): the data is very unlikely under $H_0 \rightarrow$ **reject H_0**
 - If p-value is not small (≥ 0.05): we cannot rule out $H_0 \rightarrow$ **fail to reject H_0**

*If the p-value is below **0.05**, we reject the null and conclude there **IS** an effect.*

What Is a p-Value?



p-value = probability of seeing data this extreme *if H_0 were true*

$p < 0.01$: **Strong evidence** against H_0

$p < 0.05$: **Sufficient evidence** (conventional threshold)

$p > 0.05$: **Insufficient evidence**

Example: Does Sentiment Really Affect Engagement?

- $H_0: \beta = 0$ (sentiment has **no effect** on engagement in the population)
- $H_1: \beta \neq 0$
- In our regression: $\beta = 0.042$, $n = 12,000$
- Standard error: $SE = 0.008$
- Test statistic: $t = \frac{\beta}{SE} = \frac{0.042}{0.008} = 5.25$
- p-value: **extremely small** ($p < 0.001$)

Conclusion: REJECT H_0

The positive effect of sentiment on engagement is statistically significant.

This is very unlikely to be random noise.

Two Types of Mistakes

	H_0 is TRUE (no effect)	H_0 is FALSE (effect exists)
We reject H_0	TYPE I ERROR (False Positive)	CORRECT (True Positive)
We fail to reject H_0	CORRECT (True Negative)	TYPE II ERROR (False Negative)

- **Type I Error (α):** False positive (accept 5% risk)
- **Type II Error (β):** False negative (depends on sample size)
- **Power** = $1 - \beta$ = probability of detecting a real effect

Real Consequences: Type I Error (False Positive)

Scenario: We test whether the “Product” topic gets more engagement than “Support” topics

Type I Error: We conclude Product topics get more engagement, but it is actually just noise

Consequence:

- The brand reallocates its content budget toward Product topics
- Money is wasted because there was no real difference
- The Support team loses resources for no reason

Acting on a false positive wastes budget and resources.

Real Consequences: Type II Error (False Negative)

Scenario: We test whether sentiment score predicts engagement

Type II Error: We conclude sentiment has no effect, but it actually does

Consequence:

- The brand ignores sentiment management
- Negative sentiment goes unchecked, quietly eroding engagement over months
- An opportunity to improve engagement by managing tone is missed

Missing a real effect means missed opportunities.

Statistical vs. Practical Significance (Part 1)

- With $n = 12,000$, even **tiny effects** become statistically significant
- Example: sentiment_score vs. toxicity_score: $r = 0.008$
 - With **12,000 observations**, even $r = 0.008$ might be statistically significant ($p < 0.05$)
 - But $r = 0.008$ explains only **0.006% of the variance** – practically meaningless
- This is the “**large sample trap**”: with enough data, everything becomes significant

Statistical vs. Practical Significance (Part 2)

Always ask **TWO** questions:

1. Is the effect **statistically significant**? ($p < 0.05$)
 - Is it real, or just noise?
2. Is the effect **practically meaningful**?
 - Is it large enough to matter for business decisions?

Always pair statistical significance with practical significance.

Hypothesis Testing Meets Regression

- Every coefficient in a regression output comes with:
 - The estimate (β)
 - A standard error (uncertainty in β)
 - A t-statistic (β / standard error)
 - A p-value (is β significantly different from zero?)

Predictor	β	Std. Error	t-stat	p-value	Significant?
impressions	-0.00003	0.000002	-15.0	< 0.001	Yes
sentiment_score	0.042	0.008	5.25	< 0.001	Yes
Brand: Nike	0.15	0.035	4.29	< 0.001	Yes

For each predictor, H_0 says $\beta = 0$ (no effect). If $p < 0.05$, we reject H_0 and conclude the effect is real.

Back to Our Three Questions

1. Nike's impressions vs. engagement dilemma:

- $r = -0.30$, $p < 0.001 \rightarrow$ The trade-off is **real and statistically significant**
- Implication: maximizing reach does **not** maximize engagement. Target better, not broader.

2. Apple's topic strategy:

- Topic category explains **31% of engagement variance**
- Some topics drive significantly more engagement than others (regression confirms this)
- Implication: reallocate content toward high-engagement topics

3. Coca-Cola vs. Pepsi:

- Brand explains **50% of engagement variance**
- Hypothesis test can confirm whether the specific Coca-Cola vs. Pepsi difference is significant
- Implication: brand identity is the **single biggest driver** of engagement

What You Should Remember

Five key takeaways:

1. **Variance** is the raw material – if nothing varies, there is nothing to study
2. **Correlation** tells you whether two variables move together (and how strongly), but NOT whether one causes the other
3. **Regression** draws the best-fitting line, quantifies effects, and lets you control for confounders
4. **R-squared** tells you how much of the outcome your model explains
5. **Hypothesis testing** tells you whether your findings are real or just noise – but always pair statistical significance with practical significance

The toolkit: Variance → Correlation → Regression → Hypothesis Testing

Thank You!

Dataset: Social Media Engagement Dataset
(12,000 posts, 10 brands, 5 platforms)

Tools covered: Correlation, Linear Regression, R-squared,
Confidence Intervals, Hypothesis Testing

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